

Semiconductor & Device Physics

Industry Notes — Formulas and Short Theory

A compact reference for process, device, TCAD and characterization engineers: each topic is a few lines of theory, the formulas you actually use, typical silicon numbers at 300 K, and the usual pitfalls. Notation follows Sze / Pierret / Neamen. Units: cm, cm⁻³, V, A, s unless stated (the TCAD convention); SI where noted.

Constants and material data (300 K)

Quantity	Value
q	1.602×10 ⁻¹⁹ C
k _B	1.381×10 ⁻²³ J/K = 8.617×10 ⁻⁵ eV/K
ε ₀	8.854×10 ⁻¹⁴ F/cm
V _T = kT/q	25.85 mV (300 K); ×(T/300)
kT ln10 / q	59.5 mV/decade (thermal limit of subthreshold swing)
ħ, m ₀	1.0546×10 ⁻³⁴ J·s, 9.109×10 ⁻³¹ kg

Property	Si	Ge	GaAs
Bandgap E _g (eV) (Varshni fit)	1.125	0.664	1.422
ε _r	11.7	16.2	12.9
Electron affinity χ (eV)	4.05	4.13	4.07
N _c , N _v (cm ⁻³)	2.86×10 ¹⁹ , 3.10×10 ¹⁹	1.04×10 ¹⁹ , 6.0×10 ¹⁸	4.7×10 ¹⁷ , 7.0×10 ¹⁸
n _i (cm ⁻³)	1.07e+10	2.10e+13	2.04e+06
μ _n , μ _p max (cm ² /Vs)	1360, 495	3900, 1900	8500, 400
v _{sat} n, p (cm/s)	1.07×10 ⁷ , 8.4×10 ⁶	7.0×10 ⁶ , 6.3×10 ⁶	1.2×10 ⁷ , 9.0×10 ⁶
Band type	indirect	indirect	direct (Γ)

Values are the parameter set used in the PyTCAD repository (materials.py); n_i is computed from them. Textbook tables differ by a few percent (e.g. Si n_i ≈ 1.0×10¹⁰ or 1.45×10¹⁰ depending on convention) — always state the convention.

1. Bands, carriers and statistics

Electrons occupy bands separated by the gap E_g. Conduction-band electrons (n) and valence-band holes (p) are the mobile carriers. Near the edges the density of states is parabolic, so carrier densities follow from Fermi-Dirac occupation.

Bandgap vs T	$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta}$	Varshni; Si: 1.17 eV, 4.73e-4 eV/K, 636 K
Effective DOS	$N_{c,v} = 2 \left(\frac{2\pi m_{n,p}^* kT}{h^2} \right)^{3/2} \propto T^{3/2}$	m* = DOS effective mass
Carrier density	$n = N_c \mathcal{F}_{1/2}(\eta_n)$; $\eta_n = \frac{E_{Fn} - E_c}{kT}$	Fermi-Dirac integral
Boltzmann limit	$n = N_c e^{(E_F - E_c)/kT} = n_i e^{(E_F - E_i)/kT}$ $p = N_v e^{(E_v - E_F)/kT} = n_i e^{(E_i - E_F)/kT}$	valid if E _F is >3kT below E _c
Mass action	$np = n_i^2$; $n_i = \sqrt{N_c N_v} e^{-E_g/2kT}$	equilibrium only
Intrinsic level	$E_i - E_{mid} = \frac{kT}{2} \ln \frac{N_v}{N_c}$	≈ mid-gap

NOTE Key idea

Boltzmann statistics fails for doping > ~10¹⁹ cm⁻³ (degenerate: E_F inside a band). Use Fermi-Dirac there; the error in n is exp(η)/2^{3/2}-scale and shows up in contact resistance, S/D regions and heavily doped emitters.

2. Doping, Fermi level, ionization, bandgap narrowing

Neutrality	$p - n + N_D^+ - N_A^- = 0$	full ionization: $N_D^+ = N_D$
n-type, $N_D \gg n_i$	$n \approx N_D, \quad p = \frac{n_i^2}{N_D}$	majority/minority
General	$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$	net doping $C = N_D - N_A$
Fermi level	$E_F - E_i = kT \ln \frac{n}{n_i} = -kT \ln \frac{p}{n_i}$	$\Phi_F = (E_i - E_F)/q$
Incomplete ionization	$N_D^+ = \frac{N_D}{1 + g_D e^{(E_F - E_D)/kT}}, \quad g_D = 2, \quad g_A = 4$	$E_d \approx 45 \text{ meV}$ (P, B in Si)
Bandgap narrowing	$n_{ie}^2 = n_i^2 e^{\Delta E_g/kT}$	$\Delta E_g \approx \text{tens of meV at } 10^{19}$ (Slotboom)
Resistivity	$\rho = \frac{1}{q(\mu_n n + \mu_p p)}$	$\Omega \cdot \text{cm}$; sheet $R_s = \rho/t$

Rules of thumb (Si, 300 K): $10^{16} \text{ cm}^{-3} \rightarrow \rho \approx 0.5 \Omega \cdot \text{cm}$ (n) / $1.4 \Omega \cdot \text{cm}$ (p); $10^{18} \rightarrow 0.02 / 0.04 \Omega \cdot \text{cm}$. Dopants are shallow and ~fully ionized above ~100 K up to the Mott limit ($\sim 4 \times 10^{18}$ for P).

3. Transport: drift, diffusion, mobility

Drift-diffusion	$J_n = q\mu_n n E + qD_n \nabla n, \quad J_p = q\mu_p p E - qD_p \nabla p$	total $J = J_n + J_p$
Einstein	$\frac{D}{\mu} = \frac{kT}{q} = V_T$	$D_n \approx 35 \text{ cm}^2/\text{s}$ at $\mu_n = 1350$
Matthiessen	$\frac{1}{\mu} = \frac{1}{\mu_{\text{lattice}}} + \frac{1}{\mu_{\text{imp}}} + \frac{1}{\mu_{\text{surf}}}$	scattering rates add
Caughey-Thomas	$\mu(N) = \mu_{\min} + \frac{\mu_{\max} - \mu_{\min}}{1 + (N/N_{\text{ref}})^{\alpha}}$	$N = N_A + N_D$ (total!)
Velocity saturation	$v = \frac{\mu E}{[1 + (\mu E/v_{\text{sat}})^{\beta}]^{1/\beta}}$	Canali: $\beta=2$ (e), 1 (h)
Lattice T-dependence	$\mu_{\text{lattice}} \propto T^{-3/2}$ (ideal); $T^{-2.3}$ (Si fit)	hotter $\rightarrow$ slower
Quasi-Fermi form	$J_n = -q\mu_n n \nabla \phi_n$	current $\propto$ slope of E_{Fn}
Hall	$R_H = \frac{r_H}{qp}$ (p), $-\frac{r_H}{qn}$ (n)	$r_H \approx 1.1-1.2$; gives n and $\mu_H = R_H /\rho$

NOTE Key idea

Current flows only where the quasi-Fermi level has a gradient. Drift and diffusion are each huge and nearly cancel in junctions — which is why numerical schemes need Scharfetter-Gummel fluxes (§15).

4. Generation, recombination, lifetime

SRH (mid-gap trap)	$R = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$	$n_1 = n_i e^{(E_t - E_i)/kT}$
Low-level, p-type	$R = \frac{\Delta n}{\tau_n}, \quad \tau_n = \frac{1}{\sigma v_{th} N_t}$	trap capture
Depletion region	$U = -\frac{n_i}{2\tau}, \quad J_{\text{gen}} = \frac{qn_i W}{2\tau}$	reverse leakage $\propto W$
Auger	$R = (C_n n + C_p p)(np - n_i^2)$	$C_n \approx 2.8 \times 10^{-31}, C_p \approx 9.9 \times 10^{-32} \text{ cm}^6/\text{s}$
Radiative	$R = B(np - n_i^2)$	direct gap only; $B \approx 10^{-10} \text{ cm}^3/\text{s}$ (GaAs)
Total lifetime	$\frac{1}{\tau} = \frac{1}{\tau_{\text{SRH}}} + \frac{1}{\tau_{\text{rad}}} + \frac{1}{\tau_{\text{Auger}}}$	rates add
Doping-dep. SRH	$\tau = \frac{\tau_0}{1 + N/N_{\text{ref}}}$	Scharfetter; $N_{\text{ref}} \approx 5 \times 10^{16}$
Diffusion length	$L = \sqrt{D\tau}$	Si: 10-1000 μm
Surface recomb.	$J \cdot \hat{n} = qS(n - n_0)$	S in cm/s

5. Governing equations (the semiconductor equations)

Poisson	$\nabla \cdot (\epsilon \nabla \psi) = -q(p - n + N_D^+ - N_A^-)$	ψ = electrostatic potential
Electron continuity	$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot J_n - R + G$	
Hole continuity	$\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot J_p - R + G$	
Minority diffusion eq.	$D_p \frac{d^2 \Delta p}{dx^2} - \frac{\Delta p}{\tau_p} = 0 \Rightarrow \Delta p \propto e^{-x/L_p}$	n-region, low injection
Dielectric relaxation	$\tau_d = \frac{\epsilon}{\sigma} = \rho \epsilon$	~ps in doped Si
Debye length	$L_D = \sqrt{\frac{\epsilon k T}{q^2 N}}$	Si, 10^{17} : 12.9 nm

6. P-N junction and diode

Equalising E_F leaves a depleted region of fixed ions; its field opposes diffusion. Forward bias lowers the barrier exponentially; reverse bias widens W .

Built-in voltage	$V_{bi} = V_T \ln \frac{N_A N_D}{n_i^2}$	Si $10^{17}/10^{17}$: 0.83 V
Depletion width	$W = \sqrt{\frac{2\epsilon(V_{bi} - V)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$	V forward positive
Split	$x_n = W \frac{N_D}{N_A + N_D}$, $x_p = W \frac{N_A}{N_A + N_D}$	lightly doped side takes more
Peak field	$E_{max} = \frac{q N_D x_n}{\epsilon} = \frac{2(V_{bi} - V)}{W}$	
Junction capacitance	$C_j = \frac{\epsilon A}{W} = \frac{C_{j0}}{\sqrt{1 - V/V_{bi}}}$	C-V profiling: $N = 2/(q\epsilon A^2 \cdot d(1/C^2)/dV)$
Shockley	$J = J_s (e^{V/\eta V_T} - 1)$	$\eta=1$ diffusion, $\rightarrow 2$ recombination
Saturation (long)	$J_s = q \left(\frac{D_p p_{n0}}{L_p} + \frac{D_n n_{p0}}{L_n} \right)$	short base: $L \rightarrow W_b$
Diffusion cap.	$C_d = \frac{\tau I}{V_T}$	forward bias, stored charge $Q = I\tau$
Reverse recovery	$t_s \approx \tau \ln \left(1 + \frac{I_F}{I_R} \right)$	storage time
Avalanche condition	$\int_0^W \alpha_{eff} dx = 1$, $\alpha = A e^{-B/E}$	$M = 1/(1 - \int \alpha dx) \rightarrow \infty$
Breakdown (Si, one-sided)	$BV \approx 60 \left(\frac{E_g}{1.1} \right)^{3/2} \left(\frac{N_B}{10^{16}} \right)^{-3/4} \text{ V}$	Size empirical; N_B lightly-doped side
Zener tunneling	$G_{BTBT} = A F^2 e^{-B/F}$	dominates $BV < \sim 5 \text{ V}$ (heavy doping)

NOTE Key idea

BV falls with doping (field peaks higher) and rises with E_g . Avalanche has a *positive* temperature coefficient (more phonons scatter), Zener a negative one — a quick way to identify the mechanism.

7. Bipolar junction transistor

Collector current	$I_C = I_S e^{V_{BE}/V_T}$, $I_S = \frac{qA n_i^2 D_n}{Q_G}$	Q_G = Gummel number (base dopant per area)
Current gain	$\beta = \frac{I_C}{I_B} \approx \frac{\gamma \alpha_T}{1 - \gamma \alpha_T}$	emitter efficiency γ , transport factor α_T
Emitter efficiency	$\gamma \approx \left[1 + \frac{D_p N_{AB} W_B}{D_n N_{DE} L_{pE}} \right]^{-1}$	β limited by heavy emitter + BGN
Base transport	$\alpha_T \approx 1 - \frac{W_B^2}{2L_n^2}$	
Transit time	$\tau_b = \frac{W_B^2}{2D_n}$, $f_T \approx \frac{1}{2\pi(\tau_b + \tau_e + \tau_c + \tau_{sc})}$	
Early voltage	$V_A = \frac{Q_B}{C_{BC}}$, $r_o = \frac{V_A + V_{CE}}{I_C}$	base-width modulation
Transconductance	$g_m = \frac{I_C}{V_T}$	≈ 38.7 mS per mA @ 300 K

HBT: a wide-gap emitter adds $\exp(\Delta E_g/kT)$ to β , allowing heavy base doping (low r_b) without losing gain.

8. Metal-semiconductor contacts

Barrier (n-type)	$q\phi_B = q(\Phi_M - \chi)$	Schottky-Mott ideal; real: Fermi-level pinning
Built-in potential	$V_{bi} = \phi_B - \frac{E_c - E_F}{q}$	
Thermionic emission	$J = A^* T^2 e^{-q\phi_B/kT} (e^{V/\eta V_T} - 1)$	A^* (Si-n) ≈ 112 A/cm ² K ²
Richardson constant	$A^* = \frac{4\pi q m^* k^2}{h^3} = 120 \frac{m^*}{m_0} \text{ A/cm}^2\text{K}^2$	
Image-force lowering	$\Delta\phi = \sqrt{\frac{qE_{max}}{4\pi\epsilon}}$	raises J with reverse bias
Tunneling regime	$E_{00} = \frac{q\hbar}{2} \sqrt{\frac{N}{\epsilon m^*}}$; $E_{00} \geq kT \Rightarrow$ field emission	heavy doping $\rightarrow$ ohmic
Specific contact res.	$\rho_c \propto \exp\left(\frac{2\sqrt{\epsilon m^*} \phi_B}{\hbar \sqrt{N}}\right)$ (FE)	$\Omega \cdot \text{cm}^2$; need $N > 10^{19}$

Schottky diodes are majority-carrier devices: no minority storage $\rightarrow$ fast switching, low V_F , but higher leakage than p-n diodes.

9. MOS capacitor

Gate voltage moves surface potential ψ_s : accumulation $\rightarrow$ depletion $\rightarrow$ inversion (threshold at $\psi_s = 2\Phi_F$ for the strong-inversion criterion).

Oxide capacitance	$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$	$\epsilon_{ox} = 3.9\epsilon_0$; 1 nm $\rightarrow$ 3.45 $\mu\text{F}/\text{cm}^2$
Fermi potential	$\Phi_F = V_T \ln \frac{N_A}{n_i}$	p-substrate; $10^{17} \rightarrow 0.41$ V
Flat-band voltage	$V_{FB} = \Phi_{MS} - \frac{Q_f + Q_{ot}}{C_{ox}} - \dots$	$\Phi_{MS} = \Phi_M - (\chi + E_g/2q + \Phi_F)$
Max depletion width	$W_{max} = \sqrt{\frac{2\epsilon_s(2\Phi_F)}{qN_A}}$	
Threshold voltage	$V_T = V_{FB} + 2\Phi_F + \frac{\sqrt{2q\epsilon_s N_A (2\Phi_F)}}{C_{ox}}$	long-channel, uniform doping
Series C	$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_s}$, $C_{min} = \frac{C_{ox}}{1 + \epsilon_{ox} W_{max}/(\epsilon_s t_{ox})}$	HF inversion
Body effect	$\Delta V_T = \gamma(\sqrt{2\Phi_F + V_{SB}} - \sqrt{2\Phi_F})$, $\gamma = \frac{\sqrt{2q\epsilon_s N_A}}{C_{ox}}$	
Interface traps	$C_{it} = q^2 D_{it}$	stretch-out of C-V

NOTE Key idea

QS (low-frequency) C-V rises back to C_{ox} in inversion because minority carriers follow the signal; HF C-V stays at C_{min} . The difference between the two gives D_{it} (Terman / high-low method).

10. MOSFET

Linear region	$I_D = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$	$V_{DS} < V_{GS} - V_T$
Saturation (square law)	$I_{D,sat} = \frac{\mu C_{ox} W}{2L} (V_{GS} - V_T)^2$	long channel
Velocity-saturated	$I_{D,sat} \approx W C_{ox} v_{sat} (V_{GS} - V_T)$	short channel: linear in V_{ov}
Transconductance	$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{2I_D}{V_{ov}}$	square law
Subthreshold	$I_D \propto e^{V_{GS}/(mV_T)}$, $m = 1 + \frac{C_d}{C_{ox}}$	weak inversion
Swing	$S = \ln 10 \cdot m V_T = 60 \text{ mV/dec} \cdot m$	ideal $\geq 60 \text{ mV/dec}$ at 300 K
Short-channel V_T	$\Delta V_T \propto -e^{-L/2\lambda}$, $\lambda = \sqrt{\frac{\epsilon_s t_{ox} t_{si}}{\epsilon_{ox}}}$	V_{-} roll-off; λ = natural length (thin-body)
DIBL	$\text{DIBL} = -\frac{V_T(V_{DD}) - V_T(V_{lin})}{V_{DD} - V_{lin}}$	mV/V; $\propto e^{-L/2\lambda}$
Channel length mod.	$I_D = I_{D,sat}(1 + \lambda_{CLM} V_{DS})$	output conductance g_{ds}
Unity-gain freq.	$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} \approx \frac{v_{sat}}{2\pi L}$	short channel
Intrinsic gain	$A_0 = g_m/g_{ds}$	falls with scaling

Scaling and architecture: thin-body / multi-gate devices (FinFET, GAA) shrink λ , so they tolerate shorter L at fixed leakage. Gate stack: $EOT = t_{phys} \cdot (3.9/\kappa)$; quantum confinement adds $\approx 0.3\text{--}0.5 \text{ nm}$ to the effective inversion thickness. Key parasitics: R_{sd} (contact + spreading), overlap C_{ov} , fringing C_f .

11. Leakage and reliability mechanisms

Mechanism	Dependence	Formula / rule
Junction diffusion leakage	$T^3 \exp(-E_g/kT)$	$J_s \propto n_i^2 \rightarrow$ doubles per $\sim 8\text{--}10 \text{ K}$
SRH generation leakage	$T^{3/2} \exp(-E_g/2kT)$, $\propto W$	$J = q n_i W/2\tau$
Subthreshold leakage	$\exp(V_{GS} - V_T)/(mV_T)$	I_{off} up $\approx 10\times$ per 5 mV of V_T drop
GIDL / BTBT	$\exp(-B/F)$	gate-overlap field at drain; Kane $A F^2 e^{-B/F}$
Gate tunneling	direct (thin), FN (thick)	FN: $J = C_1 E^2 \exp(-C_2/E)$; direct: WKB $T \approx \exp(-2\kappa t)$
Hot carriers (HCI)	peak near drain, substrate current I_{sub}	$I_{sub} \propto I_D \exp(-\Phi_t/q\lambda E_m)$
BTI (NBTI/PBTI)	V_T shift $\propto t^n$, $n \approx 0.15\text{--}0.25$	interface/oxide traps; recovers
TDDB	E-model / 1/E-model	lifetime $\propto \exp(-\gamma E_{ox})$ (thick) or $\exp(G/E_{ox})$
Electromigration	Black's equation	$t_{50} = A J^{-n} \exp(E_a/kT)$
Latch-up / ESD	parasitic BJT/SCR	holding V and I vs. trigger; guard rings

12. Heterojunctions, HEMT and optoelectronics

Anderson rule	$\Delta E_c = \chi_2 - \chi_1$, $\Delta E_v = \Delta E_g - \Delta E_c$	affinity rule
Thermionic over barrier	$J \propto T^2 e^{-\Delta E_c/kT}$	also tunneling through spikes
2DEG density	$n_s \approx \frac{\epsilon}{qd}(V_g - V_{th})$	HEMT; d = barrier thickness
Absorption	$\alpha(\lambda)$; $I(x) = I_0 e^{-\alpha x}$	direct gap: $\alpha \sim 10^4 \text{ cm}^{-1}$; indirect: much lower
Photocurrent	$J_{ph} = q G_{opt}(L_n + L_p + W)$	collection by diffusion + drift
Solar open-circuit	$V_{oc} = \frac{\eta kT}{q} \ln \left(\frac{J_{sc}}{J_0} + 1 \right)$	limited by J_0 (recombination)
Fill factor / eff.	$FF = \frac{P_{max}}{V_{oc} J_{sc}}$, $\eta = \frac{P_{max}}{P_{in}}$	Si cell record $\approx 26\text{--}27\%$ (practical)
LED/laser gain	$\eta_{IQE} = \frac{R_{rad}}{R_{rad} + R_{nr}}$; $\lambda_{\mu m} = \frac{1.24}{E_g(\text{eV})}$	direct-gap materials only

13. Power devices

Drift region (unipolar)	$R_{on,sp} = \frac{4BV^2}{\epsilon \mu E_c^3}$	Baliga FOM = $\epsilon \mu E_c^3$
Si limit	$R_{on,sp} \approx 5.9 \times 10^{-9} BV^{2.5} \Omega \text{ cm}^2$	BV in V
Critical field	$E_c \approx \text{Si: } 0.3, \text{ SiC: } 2.5, \text{ GaN: } 3.3 \text{ MV/cm}$	SiC/GaN $\approx 10\times$ lower R_{on} at same BV
IGBT on-state	$V_{CE(sat)} \approx V_{pn} + I R_{drift}(\text{modulated})$	conductivity modulation; tail current
Switching loss	$E_{sw} \approx \frac{1}{2} V_{DS} I_D (t_{ri} + t_{fv}) + Q_{rr} V_{DS}$	+ C_{oss} loss $\frac{1}{2} C_{oss} V^2$
Thermal	$T_j = T_a + P R_{\theta ja}$	$R_{\theta} = \Delta T/P$; $Z_{\theta}(t)$ for pulses
Safe operating area	$P_{max}(t) = \frac{T_{j,max} - T_c}{Z_{\theta}(t)}$	second breakdown / thermal runaway

14. Process physics (doping and oxide)

Implant profile	$N(x) = \frac{\Phi}{\sqrt{2\pi}\Delta R_p} \exp \left[-\frac{(x - R_p)^2}{2\Delta R_p^2} \right]$	dose Φ (cm^{-2}); LSS moments
Fick's law	$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$	$D = D_0 \exp(-E_a/kT)$
Pre-dep (const. source)	$C(x, t) = C_s \operatorname{erfc} \left(\frac{x}{2\sqrt{Dt}} \right)$	dose = $2C_s \sqrt{Dt/\pi}$
Drive-in (const. dose)	$C(x, t) = \frac{Q}{\sqrt{\pi Dt}} e^{-x^2/4Dt}$	
Junction depth	x_j : $C(x_j) = C_B$	sheet resistance $R_s = 1/(q f \mu n \text{ dx})$
Deal-Grove	$x_{ox}^2 + A x_{ox} = B(t + \tau)$	linear (B/A) $\rightarrow$ parabolic $\sqrt{Bt}$; Si consumed $0.44 x_{ox}$
Segregation	$k = \frac{C_{Si}}{C_{SiO_2}}$	B pile-down, P pile-up

Practical: D(T) is Arrhenius with $E_a \approx 3\text{--}3.7 \text{ eV}$ in Si; thermal budget $\propto Dt$. Dopant diffusion is enhanced by implant damage (TED: transient, orders-of-magnitude at low T). Channeling gives deep tails not captured by Gaussian moments.

15. TCAD numerics cheat sheet

Scaling	$\hat{\psi} = \psi/V_T, \hat{n} = n/N_s, \hat{x} = x/L_D$	N_s = peak doping
Mesh rule	$h \leq L_D$ (junctions, channels)	also resolve oxide interface & inversion layer (~1 nm)
Box (finite volume)	$\int_{V_i} \nabla \cdot J dV \rightarrow J_{i+\frac{1}{2}} - J_{i-\frac{1}{2}}$	conservative
Scharfetter-Gummel	$J_{n,k+\frac{1}{2}} = \frac{qD_n}{h} [n_{k+1}B(\delta) - n_kB(-\delta)]$	$B(x)=x/(e^x-1), \delta=\Delta\psi/V_T$
Newton	$J(u^k)\delta = -F(u^k), u^{k+1} = u^k + \delta$	coupled (ψ, n, p); quadratic
Convergence	$\max(\delta\psi , \delta n/n , \delta p/p) < \text{tol}$	update-based
Gummel	solve $\psi \rightarrow n \rightarrow p$ iteratively	linear convergence; cheap, robust at low bias
Sparse solve	LU or Krylov	LU (PARDISO/SuperLU) or Krylov + preconditioner; 3D LU memory $O(N^{4/3})$

NOTE Key idea

Typical TCAD calibration order: (1) doping profiles from SIMS, (2) mobility (CT + surface), (3) lifetimes / traps from leakage, (4) contact and series resistance, (5) high-field / quantum corrections last. Always check mesh convergence and Jacobian consistency before trusting a number.

16. Quick numbers and sanity checks (Si, 300 K)

Quantity	Typical value	Quantity	Typical value
V_T	25.85 mV	Si n_i	$\approx 1.0 \times 10^{10} \text{ cm}^{-3}$
V_{bi} ($10^{17}/10^{17}$)	0.83 V	W ($10^{17}/10^{17}, 0 \text{ V}$)	0.147 μm
Φ_F (p, 10^{17})	0.41 V	$W_{\max}$ (10^{17})	$\approx 0.10 \mu\text{m}$
Breakdown field Si	$\approx 3 \times 10^5 \text{ V/cm}$	Oxide breakdown	$\approx 10 \text{ MV/cm}$
C_{ox} (1 nm)	3.45 $\mu\text{F/cm}^2$	Diode forward drop	0.6–0.8 V
Subthreshold limit	60 mV/dec	Si thermal conductivity	1.48 W/cm·K
v_{sat} (e)	$1 \times 10^7 \text{ cm/s}$	L_D (10^{17})	12.9 nm
I_{leak} doubling	every ~8–10 K	τ_{SRH} typical	1 μs –1 ms

Checklist when a simulation looks wrong: units (cm vs μm) · doping sign convention · N vs N_{total} in mobility · mesh at junction/interface · contact type (ohmic vs Schottky) · degenerate region with Boltzmann · T consistency (n_i, V_T, μ) · statistics/BGN flags · converged tolerance on the *update* · compare against an analytic limit (V_{bi}, C_{ox}, J_s).

These notes summarize standard textbook relations (Sze & Ng; Pierret; Neamen; Baliga; Taur & Ning) for reference and are not a substitute for process-specific calibration. Empirical rules of thumb (e.g. breakdown scaling, Baliga limit coefficient, BTI exponents) vary by technology.